

Project Concept Note

EV & Freight Analytics Platform using Tableau

An interactive Business Intelligence project built on Tableau Public analysing EV infrastructure, predictive maintenance, and freight transportation

INTRODUCTION

The EV & Freight Analytics Platform is an interactive Business Intelligence project developed using Tableau Public to analyse Electric Vehicle (EV) infrastructure, predictive vehicle maintenance, and freight transportation trends. The project integrates three independent datasets and presents meaningful insights through visually interactive dashboards, enabling users to explore charging availability, battery performance and freight movement through filters and visual analytics.

PROBLEM STATEMENT

EV infrastructure data, vehicle health metrics and freight transportation records are typically scattered across separate datasets and large spreadsheets, making it difficult to see the bigger operational picture. An integrated, interactive platform is needed to transform this raw data into actionable business insights that support monitoring, maintenance planning and freight decision-making.

OBJECTIVES

- Analyse global EV charging infrastructure and station distribution.
- Monitor EV battery health and predictive maintenance indicators.
- Evaluate freight transportation patterns across commodities and trade types.
- Develop interactive dashboards for efficient data exploration.
- Present business insights through modern data visualization techniques.

DATASET DESCRIPTION

Three datasets were combined for this project:

- **EV Charging Infrastructure Dataset** - charging station locations, operators, station status, connector types, geographic coverage and installation dates
- **EV Predictive Maintenance Dataset** - battery health, State of Charge (SoC), State of Health (SoH), battery temperature, motor performance, failure probability, maintenance type, driving conditions and Remaining Useful Life (RUL)
- **Freight Transportation Dataset** - freight tons, freight value, ton-miles, trade types, commodity categories (SCTG2), transportation modes, freight growth metrics and forecast scenarios.

The datasets were cleaned and transformed using Python (Pandas & NumPy), including handling missing values, removing duplicates, standardizing categorical values, feature engineering, calculated growth metrics, date formatting and data validation.

PROPOSED DASHBOARD FEATURES

- **Dashboard 1 - EV Charging Infrastructure:** KPI cards, interactive map, tree map and bar charts covering station distribution, operational status, operator analysis, connector types and country-wise infrastructure.
- **Dashboard 2 - EV Predictive Maintenance:** KPI cards, pie chart, histogram, area chart and bar charts covering battery health, battery status, component health, failure probability, driving conditions and maintenance categories.

- **Dashboard 3 - Freight Transportation Analytics:** Highlight table (heatmap), packed bubble chart, line chart, area chart and interactive filters covering commodity analysis, trade type comparison, freight trends and transportation modes.
- **Landing Page:** A dedicated navigation dashboard providing project overview, dashboard navigation, objectives, key insights and technology stack.

EXPECTED OUTCOMES

- Simplifies interpretation of complex EV and freight data for a general audience.
- Enables meaningful comparison of EV infrastructure, vehicle health and freight trends.
- Supports operational monitoring, predictive maintenance planning and freight decision-making.
- Demonstrates practical application of BI and data visualization techniques.

KEY INSIGHTS

- The charging infrastructure spans 67 countries with over 9,700 stations, and the majority of stations are operational.
- Most vehicles maintain high battery health (approx. 88% SoH) and healthy charge levels (approx. 78% SoC), indicating good overall EV performance.
- Preventive maintenance represents the largest share of maintenance activities, while corrective and emergency maintenance occur much less frequently.
- Freight volumes vary significantly across commodity groups, with a few commodities accounting for the majority of transportation demand.

TECHNOLOGIES USED

Tableau Public, Python, Pandas, NumPy, Data Cleaning, Feature Engineering, Business Intelligence, Data Visualization.

CONCLUSION

The EV & Freight Analytics Platform successfully transforms raw EV infrastructure, predictive maintenance and freight transportation data into an interactive decision-support tool. Consisting of three integrated dashboards and one navigation page, the platform enables users to analyse EV infrastructure, predictive maintenance and freight transportation through intuitive visualizations and interactive business intelligence techniques.

Link: https://public.tableau.com/app/profile/kartik.sahani2291/viz/IBM_Project_17847153703030/Home